E-COMMERCE DATA

Analysis

Findings & Write-up

Lawrence | Data Analyst Portfolio

About the Data

Dataset

Kaggle — E-Commerce Dataset (carrie1/ecommerce-data)

Overview

This dataset contains transactional records from a UK-based online retail business operating between December 2010 and December 2011. It captures order-level data including product descriptions, quantities, unit prices, customer identifiers, and the countries orders were placed from. The dataset contains approximately 400,000 records prior to cleaning, covering both completed purchases and returned orders. Returned orders are identified by invoices prefixed with the letter 'C' and quantities that have a value of less than 0.

Key Columns

Column Name	Description
InvoiceNo	Invoice code for each order. Invoices prefixed with 'C' denote returned orders.
StockCode	Stock code for each product
Description	Product description
Quantity	Quantity of product per order
InvoiceDate	Date the order was placed
UnitPrice	Price per unit of each product
CustomerID	Unique identifier for each customer
Country	Country associated with each order

Custom Columns

Column Name	How It Was Calculated
PurchaseAmount	Quantity multiplied by UnitPrice, applied only where Quantity is greater than 0 and InvoiceNo does not begin with 'C'. All other rows are set to 0.
ReturnedAmount	Absolute value of Quantity multiplied by UnitPrice, applied only where PurchaseAmount is 0.

Data Cleaning Notes

The dataset contained a significant number of null values, though not all nulls were removed — some records remained analytically useful despite incomplete fields. The dataset also contained non-product rows such as manual entries, discount records, and postage charges, which were excluded to ensure only genuine product transactions were analyzed. Five adjustments were made:

The following adjustments were made:

1	Rows where both CustomerID and Description were null were dropped, as these records contained no usable information.
2	Rows where CustomerID was null but Description held a value were assigned the identifier 'No ID' to retain the transaction data.
3	Rows with a null UnitPrice were dropped, as purchase and return amounts could not be calculated without a price.
4	Rows where both PurchaseAmount and ReturnedAmount were 0 were dropped, as these represented transactions with no financial value.
5	Rows containing the following non-product descriptions were dropped: 'Manual', 'POSTAGE', 'Discount', 'SAMPLES', 'CRUK Commission', 'Bank Charges', 'DOTCOM POSTAGE', and 'AMAZON FEE'.

Problem Statement

Business Context

This UK-based online retail business is experiencing a high volume of returned orders, which is directly impacting total revenue. The root causes and patterns behind these returns are not yet fully understood, making it difficult for the business to take targeted corrective action. This analysis aims to investigate return patterns across products, countries, and time, while identifying where the business is generating its strongest revenue. By understanding both where revenue is being lost and where it is being made, the business can make informed decisions on where to focus its efforts to offset the impact of returns.

Questions & Findings

Q1. Return Patterns by Country and Month

FINDING	The United Kingdom accounts for the highest number of returned orders at 7,304, far exceeding the next highest country, Germany, at 436. Return volumes peak between September and November 2011, with October 2011 recording the highest monthly count at 1,122 returns, followed by November and then September. When grouped by both country and month, October 2011 in the United Kingdom records the highest combined return volume. At the product level, 'REGENCY CAKESTAND 3 TIER' has the highest number of returned orders of any individual product.

IMPLICATION	The United Kingdom's dominance in return volume is expected given that it also accounts for the largest share of total orders. However, the high return count warrants closer examination of product quality and fulfilment processes for UK-bound shipments. The concentration of returns in September through November 2011 is notable — particularly given that December 2011 recorded the lowest return volume of the period — suggesting a seasonal or operational factor specific to those three months that merits further investigation. At the product level, the return volume of 'REGENCY CAKESTAND 3 TIER' warrants a review of product quality and customer satisfaction, especially given its significance to total revenue.

Q2. Revenue by Country and Month

FINDING	The United Kingdom is the dominant revenue contributor at $8.7M, significantly ahead of the Netherlands at $283,901 and EIRE at $276,977. At the monthly level, November 2011 leads with $1.4M in total revenue, followed by October 2011 at $1.1M and September 2011 at $1.0M. When broken down by both country and month, the United Kingdom in November 2011 alone contributes $1.2M.

IMPLICATION	Despite having the highest return volume, the United Kingdom generates revenue at a scale that far exceeds every other market — making it the clear priority for operational and commercial focus. The September to November 2011 period is particularly significant: these are simultaneously the months with the highest return volumes and the highest revenue. This does not diminish the return concern, but it indicates that transaction activity was at its peak during this period. Understanding what drove the elevated return rate during these high-volume months — beyond the simple fact that more orders were placed — should be a priority for the business.

Q3. Product Purchase Volume & Revenue Contribution

FINDING	By total quantity purchased, 'PAPER CRAFT , LITTLE BIRDIE' leads at approximately 80,000 units, followed by 'MEDIUM CERAMIC TOP STORAGE JAR' at 78,000 units and 'WORLD WAR 2 GLIDERS ASSTD DESIGNS' at 55,000 units. By revenue contribution, 'REGENCY CAKESTAND 3 TIER' leads at $174,798, followed by 'PAPER CRAFT , LITTLE BIRDIE' at $168,470 and 'WHITE HANGING HEART T-LIGHT HOLDER' at $106,612.

IMPLICATION	'REGENCY CAKESTAND 3 TIER' stands out as the business's highest-value product. Despite not leading in units sold, it generates the most revenue — indicating a higher unit price relative to volume. 'PAPER CRAFT , LITTLE BIRDIE' ranks first in quantity but its revenue contribution is closely tied to that volume, which becomes a significant concern when viewed alongside its return data in Q4.

Q4. Product Return Volume & Revenue Loss

FINDING	By total units returned, 'PAPER CRAFT , LITTLE BIRDIE' leads at approximately 80,000 units, followed by 'MEDIUM CERAMIC TOP STORAGE JAR' at 74,000 units and 'ROTATING SILVER ANGELS T-LIGHT HLDR' at 9,000 units. By estimated revenue loss, 'PAPER CRAFT , LITTLE BIRDIE' accounts for $168,470, followed by 'MEDIUM CERAMIC TOP STORAGE JAR' at $77,419 and 'REGENCY CAKESTAND 3 TIER' at $9,760.

IMPLICATION	The return volume of 'PAPER CRAFT , LITTLE BIRDIE' is equal to its total purchase quantity, meaning every unit sold was subsequently returned. This renders the product a net-zero revenue contributor and indicates a fundamental issue with either product quality, customer expectations, or order fulfilment. Discontinuing or significantly reviewing this product is warranted. 'MEDIUM CERAMIC TOP STORAGE JAR' presents a similar concern — with only approximately 4,000 units successfully retained and a net revenue contribution of $4,290, its commercial viability is limited. In contrast, 'REGENCY CAKESTAND 3 TIER' sustains a revenue loss of only $9,760 against $174,798 in total revenue, reinforcing its position as the business's most valuable product.

Q5. Return Rate Trends Over Time

FINDING	Return rates across months do not follow a clear trend. January 2011 recorded the highest monthly return rate at 1.96%, followed by October 2011 at 1.88% and June 2011 at 1.84%. At the country-month level, the United Kingdom in October 2011 recorded 878 returns across 53,503 total orders, yielding a return rate of 1.64%. By contrast, the USA recorded 101 returns across only 226 total orders, producing a return rate of 44.69%. Several smaller markets recorded return rates of up to 100%, though these figures are based on very low order volumes.

IMPLICATION	The United Kingdom's return rate remaining consistently below 2% across all months is a strong indicator of healthy transaction quality at scale, particularly given that the UK processes upwards of 20,000 orders per month. Elevated return rates in low-volume markets such as the USA are statistically less meaningful given the small sample sizes involved, though they may still point to fulfilment or product-fit issues worth monitoring. No clear directional trend in return rates over time was identified — while total return counts increased in the latter months of the dataset, this appears to reflect overall order volume growth rather than a deterioration in return behaviour.

Q6. Customer Return Volume & Purchase Activity

FINDING	Customer 14911 records the highest return count at 223, followed by customer 17841 at 129 and customer 15311 at 112. Among the highest-spending customers, customer 14646 leads with $279,137 in total purchases, followed by customer 18102 at $259,665 and customer 17450 at $194,454 — all three of whom have single-digit return counts. Customer 14911, despite holding the highest return count, also records $140,878 in total purchases, placing them within the top five customers by revenue contribution.

IMPLICATION	Customer 14911 presents a unique profile — simultaneously the highest returner and a significant revenue contributor. This suggests a customer relationship worth preserving rather than deprioritising. A targeted intervention, whether through improved product recommendations, order accuracy, or direct account management, could meaningfully reduce return volume while retaining a high-value customer. The top three customers by purchase value maintaining low return counts is a positive indicator that high-value behaviour and low return behaviour tend to correlate.

Overall Recommendation

Target Segment

United Kingdom-based customers and high-return, high-value customer accounts such as customer 14911.

Reasoning

The United Kingdom accounts for the overwhelming majority of total revenue at $8.7M while simultaneously producing the highest absolute return volume. Improving transaction quality within this market represents the greatest opportunity to offset revenue loss from returns. At the individual customer level, customer 14911 exemplifies a segment where high return frequency coexists with significant purchase value — making targeted retention and return-reduction efforts commercially justified.

Suggested Actions

- Conduct a product quality review for items with the highest return volumes in the United Kingdom, prioritising 'REGENCY CAKESTAND 3 TIER' given its dual role as the top revenue contributor and most-returned product.
- Discontinue or significantly revise 'PAPER CRAFT , LITTLE BIRDIE', as its return volume equals its total purchase volume with no net revenue contribution.
- Investigate the operational and product factors that contributed to elevated return volumes during September to November 2011, particularly within the United Kingdom.
- Implement account-level monitoring for customers exhibiting high return counts alongside significant purchase activity, beginning with customer 14911.
- Establish a minimum order volume threshold before interpreting return rates for low-volume markets, to avoid misdirecting operational resources toward statistically insignificant return figures.